

where γ is any path in Ω connecting z_0 to z , and c_0 is a complex number so that $e^{c_0} = f(z_0)$. This definition is independent of the path γ since Ω is simply connected. Arguing as in the proof of Theorem 2.1, Chapter 2, we find that g is holomorphic with

$$g'(z) = \frac{f'(z)}{f(z)},$$

and a simple calculation gives

$$\frac{d}{dz} (f(z)e^{-g(z)}) = 0,$$

so that $f(z)e^{-g(z)}$ is constant. Evaluating this expression at z_0 we find $f(z_0)e^{-c_0} = 1$, so that $f(z) = e^{g(z)}$ for all $z \in \Omega$, and the proof is complete.

7 Fourier series and harmonic functions

In Chapter 4 we shall describe some interesting connections between complex function theory and Fourier analysis on the real line. The motivation for this study comes in part from the simple and direct relation between Fourier series on the circle and power series expansions of holomorphic functions in the disc, which we now investigate.

Suppose that f is holomorphic in a disc $D_R(z_0)$, so that f has a power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

that converges in that disc.

Theorem 7.1 *The coefficients of the power series expansion of f are given by*

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula gives

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

where γ is a circle of radius $0 < r < R$ centered at z_0 and with the positive orientation. Choosing $\zeta = z_0 + re^{i\theta}$ for the parametrization of this circle, we find that for $n \geq 0$

$$\begin{aligned} a_n &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(z_0 + re^{i\theta} - z_0)^{n+1}} rie^{i\theta} d\theta \\ &= \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-i(n+1)\theta} e^{i\theta} d\theta \\ &= \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta. \end{aligned}$$

Finally, even when $n < 0$, our calculation shows that we still have the identity

$$\frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since $-n > 0$, the function $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc, and by Cauchy's theorem the last integral vanishes.

The interpretation of this theorem is as follows. Consider $f(z_0 + re^{i\theta})$ as the restriction to the circle of a holomorphic function on the closure of a disc centered at z_0 with radius r . Then its Fourier coefficients vanish if $n < 0$, while those for $n \geq 0$ are equal (up to a factor of r^n) to coefficients of the power series of the holomorphic function f . The property of the vanishing of the Fourier coefficients for $n < 0$ reveals another special characteristic of holomorphic functions (and in particular their restrictions to any circle).

Next, since $a_0 = f(z_0)$, we obtain the following corollary.

Corollary 7.2 (Mean-value property) *If f is holomorphic in a disc $D_R(z_0)$, then*

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta, \quad \text{for any } 0 < r < R.$$

Taking the real parts of both sides, we obtain the following consequence.

Corollary 7.3 *If f is holomorphic in a disc $D_R(z_0)$, and $u = \operatorname{Re}(f)$, then*

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta}) d\theta, \quad \text{for any } 0 < r < R.$$

Recall that u is harmonic whenever f is holomorphic, and in fact, the above corollary is a property enjoyed by every harmonic function in the disc $D_R(z_0)$. This follows from Exercise 12 in Chapter 2, which shows that every harmonic function in a disc is the real part of a holomorphic function in that disc.

8 Exercises

1. Using Euler's formula

$$\sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i},$$

show that the complex zeros of $\sin \pi z$ are exactly at the integers, and that they are each of order 1.

Calculate the residue of $1/\sin \pi z$ at $z = n \in \mathbb{Z}$.

2. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4}.$$

Where are the poles of $1/(1+z^4)$?

3. Show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2} dx = \pi \frac{e^{-a}}{a}, \quad \text{for } a > 0.$$

4. Show that

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}, \quad \text{for all } a > 0.$$

5. Use contour integration to show that

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(1+x^2)^2} dx = \frac{\pi}{2} (1 + 2\pi|\xi|) e^{-2\pi|\xi|}$$

for all ξ real.